

Technology Stack

Date	2 July 2026
Team ID	XXXX
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end APIs that you will use to build your solution. A well-chosen technology application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for the architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	User Interface (Presentation Layer)	HTML, CSS, JavaScript	Provides an interactive and user-friendly interface for entering weather parameters and displaying flood prediction results.

2	Application Layer	Flask (Python)	Handles user requests, processes input data, connects the web interface with the machine learning model, and returns prediction results
3	Machine Learning Layer	Scikit-learn, XGBoost, Joblib	<i>[Explain why this is warranted (e.g., handling unstructured data, strong relationship mapping)]</i>
4	Cloud / Hosting Deployment (e.g., AWS, Azure, Heroku, Docker)	<i>[Specify technology]</i>	<i>[Explain why this is warranted (e.g., scalability, cost-effective hosting)]</i>
5	Version Control / CI/CD	<i>[Specify technology]</i>	<i>[Explain why this is warranted (e.g., seamless collaboration, automated testing)]</i>